

Zhanwei Zhang (张展玮)

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Education

M.Phil. in Data Science and Analytics, Hong Kong University of Science and Technology (Guangzhou) Sep 2025 ~ Present
Information Hub

Advisor: Prof. Zishuo Ding

B.Sc. in Computer Science and Technology, Southern University of Science and Technology (SUSTech) Sep 2021 ~ Jun 2025
Turing Master Class

Advisor: Prof. Yepang Liu

GPA: 3.79 / 4.0 | Weight Avg Score: 90.92 | Ranking: 36 / 195

Main courses: Mathematical Logic (A+), Java Programming (A+), Data Structures & Algorithms (A), Machine Learning (A), Database Systems (A-), Compilers (B+)

Internships

Agent Intern in Qualcomm, Shenzhen May 2026 ~ Present
Core developer of an internal AI Agent platform, evolving its architecture and enabling unified Agent / Workflow / Tool integration.

Designed Engine + Plugin architecture with dynamic backend/frontend loading via Python Entry Points and Module Federation.

Built Agent Runtime on DeepAgents, unifying Tool / MCP / Skills / Subagent integration and supporting AG-UI and HITL.

Large Language Model Intern in Lingsome, Shenzhen Aug 2024 ~ Aug 2025
Integrated multi-type Retrieval Augmented Generation (RAG) and GraphRAG systems.

Optimized GraphRAG code to extract better entities & relationships and construct domain-specific knowledge graphs.

Developed and refined pipelines to extract improved entities and relationships.

Visiting Researcher in Wuhan University, Wuhan May ~ Aug 2024
Advisor: Prof. Jinfu Chen (WHU); Prof. Weiyi Shang (UWaterloo)

Focused on software logging and failure workarounds.

Developed an automated analysis pipeline to extract, filter, and sample code commits containing try-catch blocks.

Selected Projects

OpenStar Jul 2026 ~ Present
Built Chat / Explore / Watch Agent workflows for open-source project discovery, understanding, and continuous tracking.

Designed Agent research and Watch pipelines to investigate README / Release / Activity via tools and produce structured assessments.

NIYO.AI Custom-Fit Nail 3D Modeling Platform Jan 2026 ~ Present
Engineered existing vision and geometry prototypes into a full 3D annotation and modeling platform, covering frontend, backend, database, task workflows, and the end-to-end vision-to-3D pipeline.

Built the synthetic-data and training pipeline, supporting millions of multi-view samples as well as model training, evaluation, and production inference.

Extended the parametric geometry model and rewrote its shared core in Rust with Python/WASM bindings, cutting computation from hundreds of milliseconds to ~10 ms for real-time 3D interaction.

WiTH - AI Health Companion Jun 2026
Built an AI health Agent integrating user profiles, health records, and 20+ Tool Calling capabilities for personalized health interactions. Implemented semantic memory and proactive health reminders, turning conversations into actionable tasks; won Third Prize at the Shenzhen Hackathon.

AI Micro-Drama Studio**Sep 2025 ~ Jan 2026**

Built an end-to-end pipeline that transforms novels or scripts into production-ready storyboards.

Unified creation, generation, editing, and export in one interface to streamline publish-ready video production.

Othello Game through Java and Python Programming with Strong AI**Oct ~ Dec 2021 & Mar 2023**

Developed visually appealing interface and implemented Monte Carlo & Alpha Beta Pruning algorithm.

Canteen Traffic Monitoring**Dec 2023 ~ Jan 2024**

Calculated the length of the queue by monitoring data and displayed a chart showing the changes in queue length.

Won award for finalist in National College Students' Innovation and Entrepreneurship Training program.

About 30,000 visits within three months.

Research**Numerical Error Detection in Floating-Point Computing****Sep 2024 ~ Feb 2026**

PI-detector: A condition-number-guided perturbation approach can replace costly high-precision oracles, finding 173/174 significant-error cases at about 0.13% of oracle cost (up to $73.46\times$ faster).

MGDE: Turn detection into a Newton-Raphson-guided convergent input search, reaching 80 bugs / 47 functions (vs. 70/46 ATOMU, 53/42 FPCC) while being $41.71\times$ and $10.17\times$ faster, respectively.

Reproduction and Evaluation of R1-style Reasoning Pipeline**May 2025 ~ Jun 2025**

Replicated the DeepSeek-R1 training pipeline by implementing GRPO and Cold-start SFT, significantly enhancing the multi-step reasoning capabilities of Qwen2.5 models (0.5B to 7B).

LLM-Based JSON Parser Fuzzing for Bug Discovery and Behavioral Analysis**Sep 2023 ~ Jan 2024**

Used opensource LLMs such as Llama2-7B/13B to generate test cases.

13 JSON Parsers and over 100 types of cases have been tested. Over 26 behavioral diversities have been found.

Publications

Tan, Y., **Zhang, Z.**, Ding, Z., Chen, J., & Shang, W. (2026). *ICE: Reducing search space for error-inducing input detection*. *IEEE Transactions on Software Engineering*, 1-14. [doi:10.1109/TSE.2026.3703362](https://doi.org/10.1109/TSE.2026.3703362)

Tan, Y., **Zhang, Z.**, Zhang, H., Zheng, L., Ding, Z., Chen, J., & Shang, W. (2026). *Mathematically-guided detection of floating-point errors*. *ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2026)*. [ResearchGate](#) (Accepted)

Tan, Y., **Zhang, Z.**, Chen, J., Ding, Z., Xuan, J., & Shang, W. (2025). *Computing floating-point errors by injecting perturbations*. arXiv. <https://arxiv.org/abs/2507.08467>

Han, Y., Shen, H., He, X., Mai, Z., Zhang, R., Zheng, Z., Liu, Y., Zhang, X., Li, G., **Zhang, Z.**, Liang, Z., Chen, Y., Xie, Y., Li, M., Shen, G., Wang, C., Ye, J., Zhu, L., Fu, T.-M., & Yang, X. (2025). A comprehensive analysis of interflight variability in carbon dioxide emissions from global aviation. *Environmental Science & Technology*, 59(12), 6179-6191. doi:10.1021/acs.est.5c02371

Patents**一种点餐方法、系统、终端及介质 (Innovative Ordering Method, System, Terminal, and Medium Patent)****May 2023**

Innovated a method and system to alleviate peak-hour traffic in cafeterias.

Applied on May 5, 2023; Application no: 202310498065

Skills

Languages: English (Fluent; IELTS: 6.5), Mandarin (Native), Cantonese (Native)

Programming Languages & Frameworks: Java, Python, C/C++, SQL, Spring Boot, Vue, React

Tools: IntelliJ IDEA, PyCharm, Visual Studio Code, Anaconda, Git, CMake

Honors & Scholarships

Third Prize, Shenzhen Hackathon

Jun 2026

Special Innovation Award (Unique Winner) & Second Prize, OpenHarmony Competition Training Camp

Sep 2025

Postgraduate Studentship (PGS), HKUST(GZ)	Sep 2025
Outstanding Student, SUSTech	Jan 2024
Honorable Mention, Mathematical Contest in Modeling	May 2023
Finalist, National College Students' Innovation and Entrepreneurship Training program	Jun 2023
Third Prize, China Undergraduate Mathematical Contest in Model	Sep 2023